

WARROAD, MN, USA

WMO: 726548

Lat: 48.941N

Lon: 95.348W

Elev: 327

StdP: 97.45

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.9	-35.9	0.1	-30.5	-32.4	0.2	-28.0	10.1	-10.3	9.0	-9.9	2.1	260	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.0	22.5	27.7	21.3	26.3	20.0	23.9	27.8	22.6	26.5	21.4	25.3	4.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.9	26.6	21.3	16.6	25.5	19.8	15.1	24.3	73.6	28.0	68.2	26.5	63.3	25.6	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.6	-34.8	32.1	3.5	1.6	-37.3	33.3	-39.3	34.2	-41.2	35.1	-43.8	36.2		
(5)			-34.8	25.9	3.3	1.9	-37.2	27.3	-39.1	28.4	-41.0	29.5	-43.3	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-14.5	-12.5	-4.3	3.8	10.9	17.0	19.7	18.3	13.7	6.1	-2.2	-10.6		
	DBStd	13.24	8.36	7.64	7.32	5.50	4.86	3.63	3.19	3.44	4.63	5.04	6.18	7.48		
	HDD10.0	3239	758	631	444	196	48	1	0	1	16	141	367	638		
	HDD18.3	5427	1016	864	702	436	234	67	23	43	151	378	616	897		
	CDD10.0	1003	0	0	1	10	76	210	300	257	127	21	1	0		
	CDD18.3	148	0	0	0	0	4	26	64	41	12	1	0	0		
	CDH23.3	1237	0	0	0	2	46	222	526	333	105	4	0	0		
	CDH26.7	272	0	0	0	0	7	51	119	73	22	1	0	0		
(14) Wind	WSAvg	3.5	3.4	3.4	3.5	3.9	4.0	3.6	3.2	3.1	3.5	3.8	3.8	3.4		
(15) Precipitation	PrecAvg	644	20	18	22	36	82	119	92	83	73	54	29	23		
	PrecMax	851	45	54	74	79	193	323	143	168	199	133	99	41		
	PrecMin	441	5	0	5	8	35	48	43	23	15	8	5	10		
	PrecStd	108	10	13	17	16	42	65	31	35	47	33	23	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	6.1	16.3	22.5	27.4	31.1	32.0	31.2	29.0	22.9	15.0	4.9		
		MCWB	2.3	3.8	10.8	13.6	17.5	22.8	24.1	24.4	21.7	16.2	10.3	2.4		
	2%	DB	2.0	2.8	10.1	17.9	24.0	27.8	29.1	28.6	26.4	19.1	11.1	2.6		
		MCWB	0.7	1.5	6.9	10.8	16.0	20.8	22.7	22.9	19.7	14.4	8.0	1.1		
	5%	DB	-0.9	1.3	7.2	15.9	22.1	26.1	27.7	27.2	23.7	17.0	7.9	1.2		
		MCWB	-1.5	0.1	4.4	9.5	14.9	19.3	21.8	21.1	18.2	12.5	5.6	-0.1		
	10%	DB	-2.7	-1.1	4.2	12.5	19.0	23.8	26.4	25.1	22.0	13.7	6.1	-0.9		
		MCWB	-3.6	-1.7	2.2	7.5	13.6	18.3	20.5	19.5	17.3	10.4	3.8	-1.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.9	11.8	14.2	20.1	24.5	25.7	25.6	23.4	17.7	11.2	3.0		
		MCDB	4.5	5.3	15.5	19.6	24.9	28.1	29.4	29.9	28.2	21.0	13.9	4.4		
	2%	WB	0.2	2.0	7.0	11.6	17.7	22.3	24.2	23.6	20.8	15.0	8.0	1.3		
		MCDB	1.3	3.0	9.6	17.3	21.9	27.0	27.9	27.4	24.8	19.0	9.9	2.6		
	5%	WB	-1.6	0.1	4.4	9.9	15.7	20.3	22.9	21.9	19.0	12.6	5.9	0.0		
		MCDB	-1.1	1.2	6.9	14.9	19.6	24.4	26.4	25.9	23.0	16.0	8.2	1.0		
	10%	WB	-3.7	-1.9	2.7	8.0	14.4	19.0	21.7	20.5	17.6	10.5	3.8	-1.6		
		MCDB	-2.7	-1.4	4.7	12.0	18.9	23.0	25.5	24.2	20.8	13.5	5.9	-1.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.4	10.0	10.2	10.5	10.0	10.6	10.7	10.5	8.7	7.5	8.0		
		MCDBR	9.1	9.2	10.8	14.8	14.3	12.7	12.0	12.6	13.4	12.8	10.0	8.8		
		MCWBR	8.3	7.8	8.1	9.3	8.1	7.5	7.3	7.7	8.2	8.4	7.5	7.7		
		MCDBR	9.2	8.4	10.5	13.7	11.7	11.1	10.7	11.3	11.9	11.6	9.6	8.5		
	5% WB	MCWBR	8.5	7.3	8.0	9.1	7.6	7.3	7.3	7.7	8.0	8.2	7.3	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.293	0.294	0.344	0.360	0.376	0.380	0.374	0.346	0.304	0.275	0.254		
	taud		2.292	2.200	2.398	2.382	2.401	2.383	2.390	2.417	2.483	2.568	2.486	2.357		
	Ebn at Noon		807	850	926	905	900	884	874	864	857	849	798	770		
	Edh at Noon		70	99	97	110	113	116	113	105	89	69	59	58		
(44) All-Sky Solar Radiation	RadAvg		1.37	2.57	3.76	4.78	5.38	5.65	6.03	5.03	3.71	2.18	1.22	0.98		
	RadStd		0.15	0.20	0.37	0.43	0.35	0.45	0.29	0.37	0.30	0.32	0.14	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page